

BARABINSK, Russian Federation

WMO: 296120

Lat: 55.3545N

Lon: 78.3078E

Elev: 120

StdP: 99.90

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.7	-32.7	-38.8	0.1	-35.5	-35.9	0.1	-32.5	9.4	-9.3	8.7	-8.7	1.7	30	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	29.3	18.8	27.7	18.2	26.1	17.5	20.5	26.4	19.5	25.4	18.6	24.2	3.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.5	23.2	17.5	12.7	22.0	16.5	11.9	21.2	59.9	26.5	56.4	25.4	53.4	24.1	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.9	7.0	DB	-37.7	31.9	3.9	1.7	-40.6	33.2	-42.9	34.2	-45.1	35.1	-47.9	36.3	(4)
(5)			WB	-37.8	22.6	3.9	2.0	-40.6	24.0	-42.9	25.2	-45.1	26.3	-47.9	27.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-18.8	-15.7	-7.4	3.5	11.7	17.2	18.9	16.2	10.1	3.0	-7.4	-14.9	(6)	
	DBStd	14.47	8.12	7.23	6.08	5.80	5.52	4.34	3.29	3.75	4.54	4.95	7.80	8.16	(7)	
	HDD10.0	3981	892	718	540	208	48	4	0	2	54	222	523	771	(8)	
	HDD18.3	6278	1151	952	798	446	216	71	34	86	251	474	773	1029	(9)	
	CDD10.0	864	0	0	0	12	100	219	276	195	56	6	0	0	(10)	
	CDD18.3	119	0	0	0	0	9	36	51	21	2	0	0	0	(11)	
	CDH23.3	1154	0	0	0	2	126	353	438	208	28	0	0	0	(12)	
	CDH26.7	261	0	0	0	0	27	85	103	42	4	0	0	0	(13)	
(14) Wind	WSAvg	3.3	3.2	3.4	3.6	3.8	3.7	3.0	2.8	2.8	3.1	3.3	3.6	3.6	(14)	
(15) Precipitation	PrecAvg	377	20	14	13	20	30	48	60	51	37	32	28	22	(15)	
	PrecMax	563	50	28	50	60	73	155	148	104	87	65	59	62	(16)	
	PrecMin	211	2	2	2	2	8	6	8	9	0	9	8	2	(17)	
	PrecStd	71	11	7	9	12	18	32	38	24	22	14	12	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	0.8	5.9	22.4	29.4	31.0	31.6	30.4	26.3	19.1	7.3	1.2	(19)	
		MCWB	-0.7	-0.2	2.6	12.4	15.9	19.2	20.8	18.9	15.8	10.7	5.5	0.2	(20)	
	2%	DB	-2.4	-0.9	3.6	18.6	26.6	29.1	29.2	27.8	23.2	15.5	5.1	0.0	(21)	
		MCWB	-3.1	-1.9	1.3	10.9	15.0	18.4	19.6	18.6	14.4	9.6	3.7	-0.6	(22)	
	5%	DB	-4.4	-2.9	2.1	15.5	24.3	27.2	27.5	25.6	20.2	12.6	3.3	-1.8	(23)	
		MCWB	-5.1	-3.7	0.5	9.4	14.0	17.8	18.9	17.4	13.3	8.0	2.0	-2.5	(24)	
	10%	DB	-6.6	-5.1	0.9	12.5	21.4	25.2	25.8	23.8	17.7	10.1	1.3	-4.0	(25)	
		MCWB	-7.2	-5.8	-0.5	7.5	12.6	17.0	18.5	16.8	11.9	6.9	0.3	-4.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.5	-0.1	3.1	13.6	17.6	21.2	22.3	20.8	17.2	11.9	5.9	0.5	(27)	
		MCDB	0.2	0.7	5.3	20.4	26.9	28.3	28.6	26.5	23.5	17.2	6.7	1.0	(28)	
	2%	WB	-3.0	-1.8	1.6	11.5	16.0	19.6	21.0	19.6	15.2	10.2	3.8	-0.7	(29)	
		MCDB	-2.3	-1.0	3.2	17.5	25.0	26.2	26.6	25.6	21.7	14.0	5.0	0.0	(30)	
	5%	WB	-5.1	-3.7	0.6	9.7	14.6	18.6	20.1	18.5	13.9	8.6	2.1	-2.5	(31)	
		MCDB	-4.5	-2.9	2.0	15.1	22.6	25.1	25.7	23.9	19.1	12.3	3.2	-1.9	(32)	
	10%	WB	-7.2	-5.8	-0.4	7.8	13.3	17.8	19.2	17.4	12.5	6.9	0.5	-4.6	(33)	
		MCDB	-6.6	-5.1	0.9	12.3	20.0	23.9	24.4	22.4	17.1	9.7	1.2	-4.0	(34)	
(35) Mean Daily Temperature Range	MDBR		7.6	8.3	8.6	9.1	11.0	10.3	10.0	10.5	9.7	7.4	6.2	7.3	(35)	
	5% DB	MCDBR	8.3	7.2	7.8	13.4	14.5	13.4	12.8	13.6	13.8	12.0	5.6	7.4	(36)	
		MCWBR	8.3	7.0	6.4	7.6	6.1	5.4	5.1	5.8	6.7	6.9	4.8	7.1	(37)	
	5% WB	MCDBR	8.3	7.2	6.9	12.8	13.2	11.7	11.2	11.4	12.5	10.8	5.5	7.5	(38)	
		MCWBR	8.3	7.1	5.9	7.6	6.3	5.3	5.3	5.5	6.5	6.8	4.9	7.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.286	0.317	0.373	0.408	0.402	0.408	0.434	0.416	0.365	0.351	0.319	0.288	(40)	
	taud		2.024	1.992	1.921	2.099	2.228	2.285	2.207	2.233	2.333	2.231	2.075	1.973	(41)	
	Ebn at Noon		593	719	763	806	839	838	804	793	788	688	543	478	(42)	
	Edh at Noon		68	103	142	138	129	123	131	120	95	82	65	57	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.93	1.98	3.49	4.61	5.45	5.93	5.39	4.39	3.05	1.52	0.72	0.59	(44)	
	RadStd		0.07	0.11	0.18	0.36	0.47	0.55	0.36	0.39	0.30	0.17	0.13	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page